

MADRE

Agentic
System

• MODEL-AGNOSTIC • DELAYED • REASONING • EFFORT • [Beta v0.1]

Operations Guide And Glossary

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UX, Observability, And Recovery

1.1 Interaction Contract

State	User-visible meaning
Immediate answer	The foreground has enough safe local context to answer.
Clarification needed	Intent, scope, risk, or context is insufficient.
Queued reasoning	Work has been recorded for delayed execution.
Running task	Delayed reasoning or action work is active.
Blocked by policy	A policy boundary prevents execution or promotion.
Authorization required	The action crosses a configured authority boundary.
Failed task	Work ended without acceptable result or recovery.
Completed task	Result is available with <code>RuntimeJournal</code> trace evidence.
Corrected or rolled back	A previous result or knowledge-boundary change has been revised or reverted.
Invalidated or superseded	Prior evidence, source, or result is no longer active because it was corrected, revoked, reclassified, or replaced.
Deleted or detached	Active generated material is removed from use where allowed while accountable <code>RuntimeEvent</code> history remains distinguishable.
Learning candidate	An evaluated <code>LearningCandidate</code> proposal exists but is not active.

1.2 Observability Surfaces

The local inspection surface must expose sessions, `ReasoningTask`, `RuntimeEvent`, context bundles, `RuntimeEvent` evidence and generated material, internal action evidence, `PolicyDecision`, failures, metrics, checkpoints, rollbacks, invalidations, supersessions, deleted/detached active material, `KnowledgeCandidate`, `LearningCandidate`, and recovery actions. The inspection surface is not a privileged bypass; it is a read-first view over the same policy-governed runtime.

Governance, Deployment, And Limits

2.1 Secure Engineering Principles

MADRE requires secure defaults for local credentials, dependency selection, governed external boundaries, secrets handling, reviewable release records, and vulnerability response. Security evidence must cover both conventional software risks and model-mediated risks such as prompt injection, poisoned retrieval, unsafe output handling, excessive agency, and remote disclosure.

2.2 Open-Source Governance

MADRE is released under GPL-3.0. The open-source governance model must preserve license clarity, dependency review, contribution rules, security disclosure, release integrity, and community responsibility. Openness is part of the project value only when users and developers can inspect, adapt, and verify the runtime without surrendering private data to a platform owner.

2.3 Deployment And Operational Assumptions

The baseline deployment target is an ordinary local device with local storage, local configuration, optional local model execution, and optional remote integrations governed by policy. The operational profile includes storage paths, model setup assumptions, backup and restore, migration, diagnostics, log retention, offline behavior, and upgrade or rollback expectations.

Deployment assumptions remain conceptual at this level of specification. The product architecture defines authority, locality, inspection, recovery, model-use boundaries, and action boundaries without reducing deployment to a single operational shape.

2.4 Risks, Limitations, And Excluded Functions

MADRE limits active executable behavior by authority class. Workflow authoring, self-learning, speech interaction, graphical product surfaces, distributed inference, organization-scale deployment, and model fine-tuning remain governed by the same local-first, policy-governed, model-agnostic, auditable, and recoverable runtime boundary as the rest of the product.

Glossary

Term	Definition
AgentAction	Smallest executable capability available under module or agent authority.
AgentRequest	Concrete invocation envelope constructed for an agent, routine, action, or model-backed action.
AgentResponse	Concrete result envelope returned by an agent, routine, or action.
AgentRoutine	Agent-owned compound executable capability that receives AgentRequest and returns AgentResponse .
Boundary composition	Runtime construction of a valid path from accumulated module, agent, workflow, routine, action, model-use, context, storage, privacy, and learning constraints.
Core Module	Kernel-assigned role held by one compatible ReasoningModule for ordinary interaction fallback; it is not a module subtype.
ComplexAgent	MADREAgent subtype capable of foreground continuity, deeper reasoning supervision, and reflection without gaining extra authority.
ContextBundle	Minimized selected context for a specific runtime purpose; inclusion does not make source material instruction or authority.
KnowledgeCandidate	Pending change to a module knowledge boundary, not active knowledge and not learning.
KnowledgeRecord	Broad uncertain retained module knowledge with provenance, scope, sensitivity, intended use, truth authority, confidence, correction, and lifecycle metadata.

Term	Definition
LearningCandidate	Evaluated proposed module improvement that remains inactive until promoted through the learning boundary.
MADREKernel	Deterministic local coordinator for module registration, target resolution, Core Module assignment, scheduling, runtime services, and event append.
MADREAgent	Abstract module-owned runtime actor that executes concrete invocations under module boundaries.
Main Agent	Role held by one module-owned SystemAgent ; it is not a separate class and grants no cross-module authority.
ModelAction	AgentAction specialization that uses an allowed ModelBinding and returns ordinary produced material with RuntimeEvent evidence.
ModelBinding	Configuration boundary constraining model-backed execution options without becoming the model or authority.
PolicyDecision	Passive record of an allow, block, or authorization-required boundary outcome.
ReasoningArtifact	Produced or materialized output; not knowledge, truth, policy, authority, or learning by default.
ReasoningModule	Governed reasoning and data-governance unit that owns module knowledge, agents, workflows, routines, actions, context governance, and local boundaries.
ReasoningPlan	Durable reasoning-continuity object containing planned task occurrences.
ReasoningRequest	Passive module-level reasoning or delegation need.

Term	Definition
ReasoningResponse	Passive module-level response representing answer, clarification, block, delegation, plan-created, or failure material.
ReasoningTask	Planned execution occurrence inside a ReasoningPlan ; concrete AgentRequest is built later at execution selection.
RuntimeEvent	Append-only runtime fact or evidence record.
RuntimeJournal	Append-only local trace mechanism for runtime events and inspection payload references.
SimpleAgent	Direct-execution MADREAgent subtype.
SystemAgent	Module-bound ComplexAgent used for ordinary module operation and Main Agent role assignment.
TwinAgent	MADREAgent subtype justified by mandatory checked-output behavior.
Workflow	Module-owned planning blueprint that creates or refines ReasoningTask occurrences; not executable work.
Read-first console	Local inspection surface that exposes status and trace without becoming a privileged mutation path.
Recovery path	Defined correction, rollback, deletion, compensation, blocking, invalidation, supersession, or detachment behavior for an outcome or state transition.